

AROMA

AI-powered Repository & Object Model Analyzer

Architecture Deep Dive

Full-stack TypeScript with IBM watsonx.ai

IBM Bob Hackathon 2026 - Turn idea into impact faster
Built with IBM Bob IDE | Powered by IBM watsonx.ai Granite

System Architecture

AROMA is a full-stack web application with a React frontend (Next.js App Router) and five server-side API routes that call IBM watsonx.ai. All IBM credentials live exclusively on the server - never sent to the browser.

Core Components

Frontend:	Next.js 15 App Router + React 19 + Tailwind CSS 4
State:	Zustand 5 global state store
UI Library:	shadcn/ui (30+ accessible components)
Charts:	Recharts (6 chart types in Health Dashboard)
Backend:	5 Next.js API Routes (server-side only)
AI Client:	lib/watsonx.ts (IBM watsonx.ai REST + IAM)
Database:	Prisma ORM + SQLite

Data Flow

1. User pastes code in browser
2. Frontend calls POST /api/* with code payload
3. API route builds system prompt + user message
4. lib/watsonx.ts obtains IAM token (cached, 60-min TTL)
5. POST to IBM watsonx.ai /ml/v1/text/chat endpoint
6. Granite model returns structured JSON or Markdown
7. API route parses response, applies fallback if needed
8. Frontend renders results in UI module

Security Architecture

- All AI calls server-side only - API keys never reach browser
- .env is gitignored and excluded via .bobignore
- Input validation on all routes (code/message required)
- TypeScript strict mode enabled
- Fallback data for every route when AI is unavailable